

Advanced Programming Practices

MTech 2026

Lab - 2

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP1_Qn.c (e.g. ABC2026001_APP1_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a program to reverse an array of integers without using any additional array.
2. Take a 3-dimensional array, say $a[p][r][c]$, where p denotes planes, each with r rows and c columns. Initialize the array with numbers sequentially from 1 to $p*r*c$, and take a pointer, x , pointing to the base address of the array. Write a program to print the last element of each plane accessed through the pointer.
3. Write a program to create an array with n elements. For the user-given input, x , rotate the array elements to the left with x positions.
For example, array $a[] = \{1, 2, 3, 4\}$ and $x = 2$, then output should be $a[] = \{3, 4, 1, 2\}$ and for $x = 5$, output should be $a[] = \{2, 3, 4, 1\}$.
4. Given an array of n integers, that may contain repeating elements. Write a C program to print the unique elements along with the frequency of that element in the input array.
5. Given an array of integers, write a C program to find the index where the sum of elements at the left-hand side of it is the same as the sum of elements at the right-hand side of it.

For example: input array : -7 1 5 2 -4 3 0 , output: index = 3